

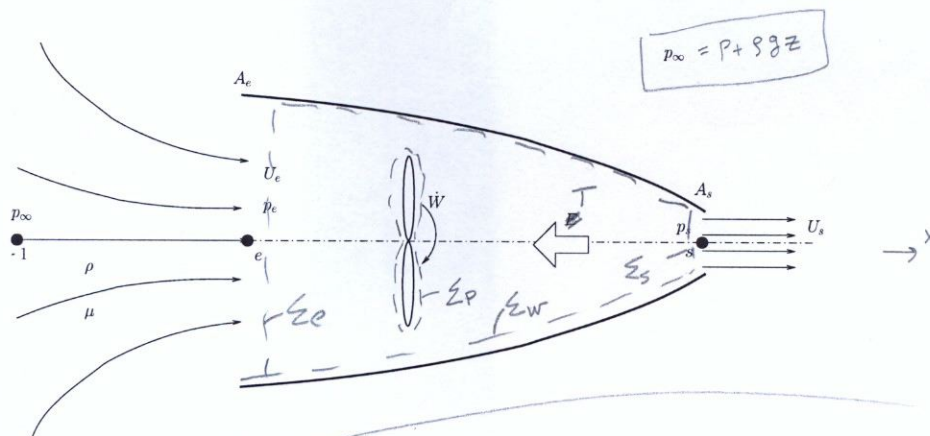
CL 9

The submarine propulsion device schematically represented in the figure consists of a nozzle, with inlet and outlet crosssection area A_e and A_s , and a ~~fan~~^{propeller} located inside, which imparts a rate of work per unit time (power) $\dot{W}$ to the circulating water stream, whose density and viscosity are ρ and μ , respectively. To determine the thrust T produced when the submarine is at rest and at sufficient depth for gravity to have a negligible effect, please follow these steps:

1. By application of the continuity equation determine a relationship between the uniform velocities U_e and U_s at the inlet and outlet sections.
2. Use the momentum conservation equation to obtain an expression for the thrust (axial force) that the fluid exerts on the propulsion device as a function of the flow conditions at the inlet and outlet. Assume that the pressure surrounding the device is equal to the ambient pressure p_∞ .
3. Assuming that the flow upstream from the entrance is steady and inviscid, use Bernoulli's equation along a given streamline to find a relationship between the flow conditions in the stagnant atmosphere (1) and at the entrance e .
4. Use the energy equation to relate the work per unit time $\dot{W}$ with the different fluid variables appearing in the problem. Assume that the variation of internal energy between the inlet and outlet sections is negligible ($e_e = e_s$) and that all of the device walls, including those of the fan, are thermally insulated.
5. Combine the above equations to show that the thrust can be expressed in the form

$$T/(\rho^{1/3} \dot{W}^{2/3} A_s^{1/3}) = f(A_e/A_s),$$

obtaining the function f .



$$P = P_0 + \rho g z$$

$$-\int_{\Sigma_c} p \bar{n} d\sigma + \int_{V_c} \rho \bar{g} dV = -\int_{\Sigma_c} (P + \rho g z) \bar{n} d\sigma$$

$$\textcircled{1} U_e A_e = U_s A_s$$

$$\textcircled{2} \bar{e}_x \left(\rho U_s^2 A_s - \rho U_e^2 A_e \right) = - \int_{\Sigma_e + \Sigma_s + \Sigma_w + \Sigma_p} (P - P_\infty) \bar{n} d\sigma + \int_{\Sigma_w + \Sigma_p} \bar{e}_x \cdot \bar{n} d\sigma = (P_e - P_\infty) A_e \bar{e}_x - \bar{F}$$

$$\textcircled{3} P + \rho g z + \frac{1}{2} \rho U^2 = P + \frac{1}{2} \rho U^2 = P_\infty = P_e + \frac{1}{2} \rho U_e^2$$

$$\bar{F} = -\rho U_e A_e (U_s - U_e) \bar{e}_x + (P_e - P_\infty) A_e \bar{e}_x$$

$$\textcircled{4} \rho U_e A_e \left[\left(\frac{P_e}{\rho} + \frac{1}{2} U_s^2 \right) - \left(\frac{P_e}{\rho} + \frac{1}{2} U_e^2 \right) \right] = - \int_{\Sigma_e + \Sigma_s + \Sigma_w + \Sigma_p} \bar{v} \cdot \bar{n} d\sigma + \int_{\Sigma_e + \Sigma_s + \Sigma_w + \Sigma_p} \bar{v} \cdot \bar{e}_x \bar{n} d\sigma + \int_{\Sigma_e + \Sigma_s + \Sigma_w + \Sigma_p} \kappa \bar{v} \cdot \bar{n} d\sigma = P_e U_e A_e - P_s U_s A_s + \dot{W}$$

$$\dot{W} = \rho U_e A_e \left[\left(\frac{P_s}{\rho} + \frac{1}{2} U_s^2 \right) - \left(\frac{P_e}{\rho} + \frac{1}{2} U_e^2 \right) \right] = \rho U_e A_e \left[\left(\frac{P_\infty}{\rho} + \frac{1}{2} U_s^2 \right) - \frac{P_e}{\rho} \right] = \frac{1}{2} U_s^2 \rho U_e A_e = \frac{1}{2} \rho U_s^3 A_s \rightarrow U_s = \left(\frac{2 \dot{W}}{\rho A_s} \right)^{1/3}$$

$$\textcircled{5} T = \rho U_e A_e (U_s - U_e) - (P_e - P_\infty) A_e = \rho U_s A_s (U_s - U_e) = \rho A_s U_s^2 \left(1 - \frac{1}{2} \frac{A_s}{A_e} \right) = \left(\rho A_s \dot{W}^2 \right)^{1/3} \frac{2^{2/3}}{1 - \frac{1}{2} \frac{A_s}{A_e}} = f$$